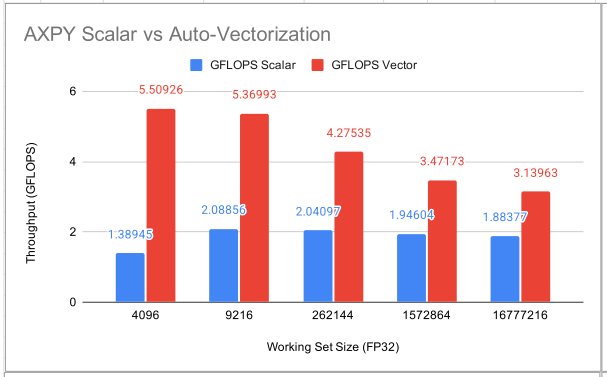
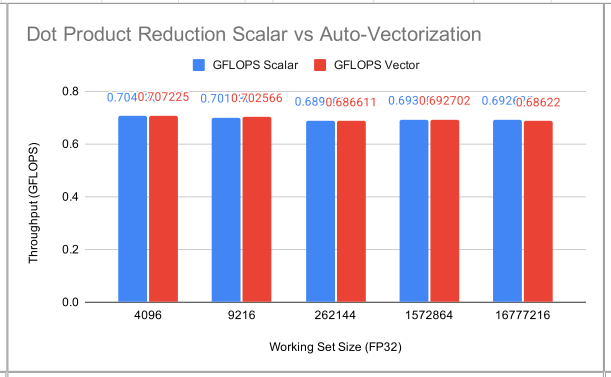
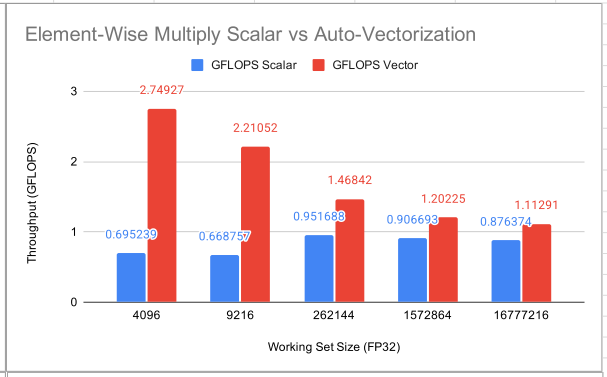
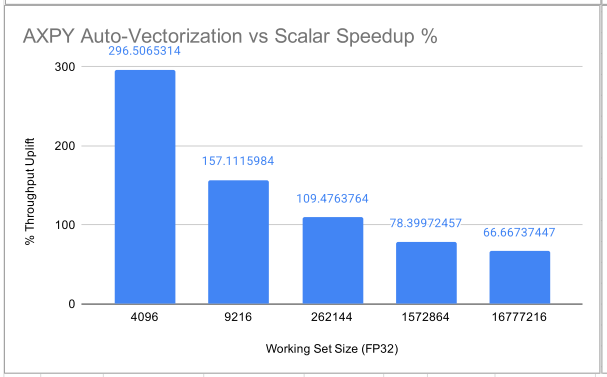
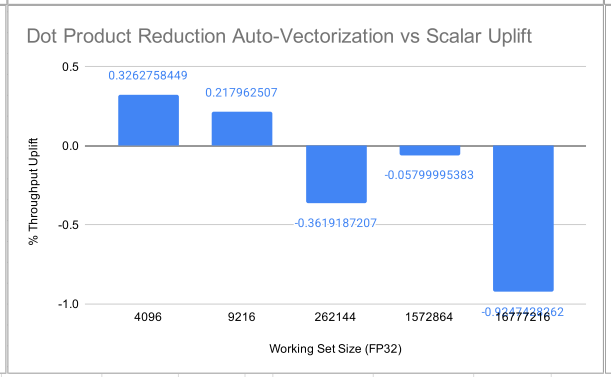
AXPY-->	Working set:	GFLOPS Scalar	GFLOPS Vector	Speedup %	DotProduction-->	Working set:	GFLOPS Scalar	GFLOPS Vector	Speedup %	Working set:	ElementWiseMultiply-->	GFLOPS Scalar	GFLOPS Vector	Speedup %
L1-I	4096	1.38945	5.50926	296.5065314		4096	0.704925	0.707225	0.326275844	4096		0.695239	2.74927	295.4424306
L1-D	9216	2.08856	5.36993	157.1115984		9216	0.701038	0.702566	0.217962507	9216		0.668757	2.21052	230.5415869
L2	262144	2.04097	4.27535	109.4763764		262144	0.689105	0.686611	-0.36191872	262144		0.951688	1.46842	54.29636604
L3	1572864	1.94604	3.47173	78.39972457		1572864	0.693104	0.692702	-0.05799995	1572864		0.906693	1.20225	32.59725177
DRAM	16777216	1.88377	3.13963	66.66737447		16777216	0.692625	0.68622	-0.92474282	16777216		0.876374	1.11291	26.99030323

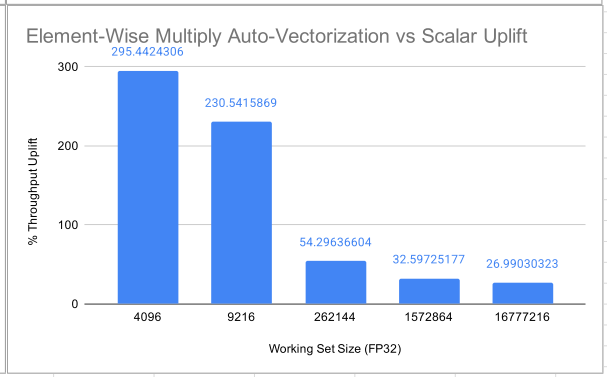




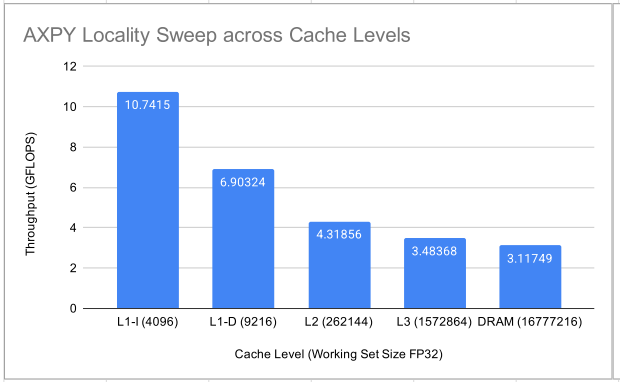


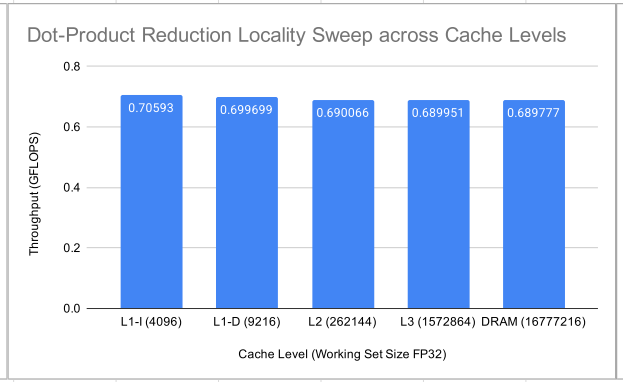


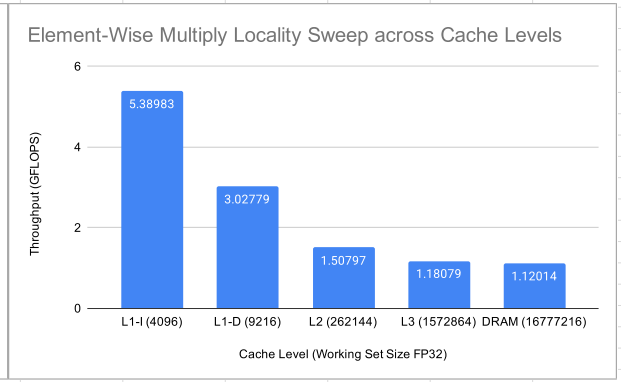




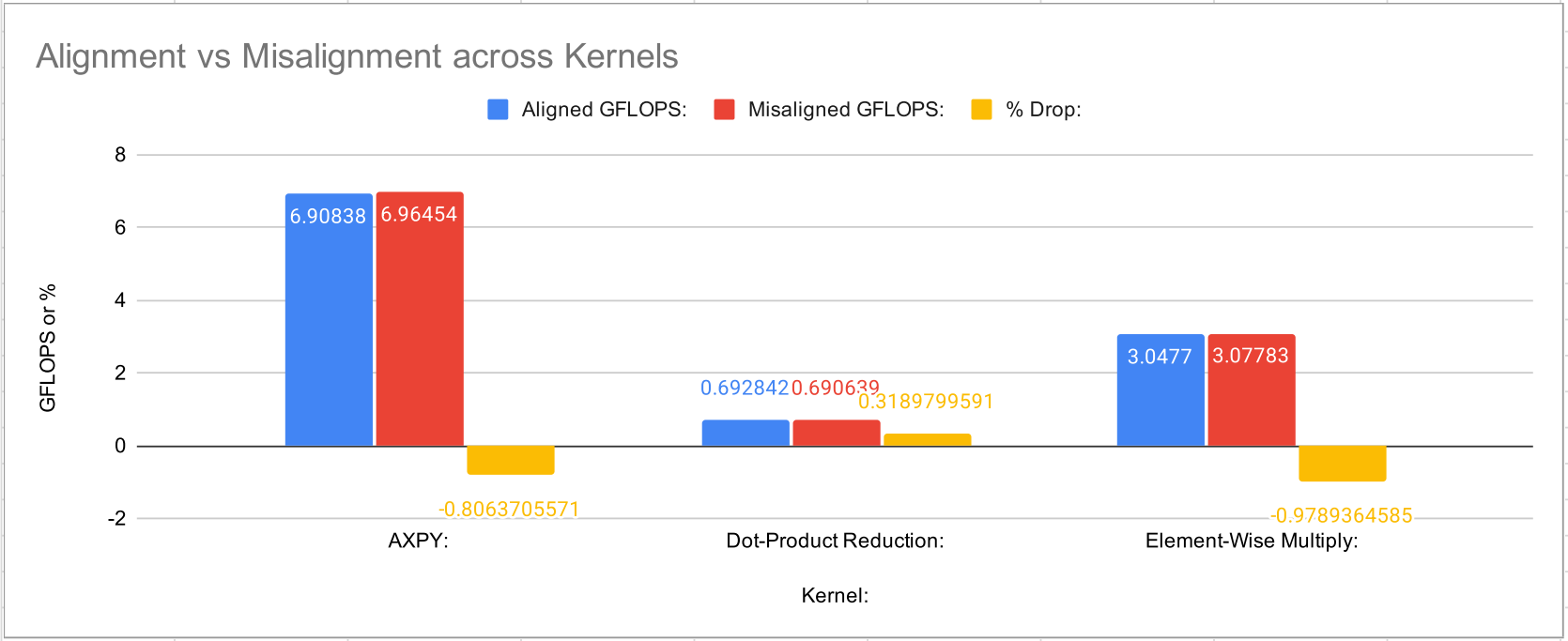
AXPY:	Working Set:	HW Arithmetic Intensity (FLOPS/Byte):	GFLOPS:	Dot-Product Reduction:	HW Arithmetic Intensity (FLOPS/Byte):	GFLOPS:	Element-Wise Multiplication:	HW Arithmetic Intensity (FLOPS/Byte):	GFLOPS:
L1-I (4096)	4096	0.166667	10.7415		0.249998	0.70593		0.0833333	5.38983
L1-D (9216)	9216	0.166667	6.90324		0.2499999	0.699699		0.0833333	3.02779
L2 (262144)	262144	0.166667	4.31856		0.25	0.690066		0.0833333	1.50797
L3 (1572864)	1572864	0.166667	3.48368		0.25	0.689951		0.0833333	1.18079
DRAM (16777216)	16777216	0.166667	3.11749		0.25	0.689777		0.0833333	1.12014



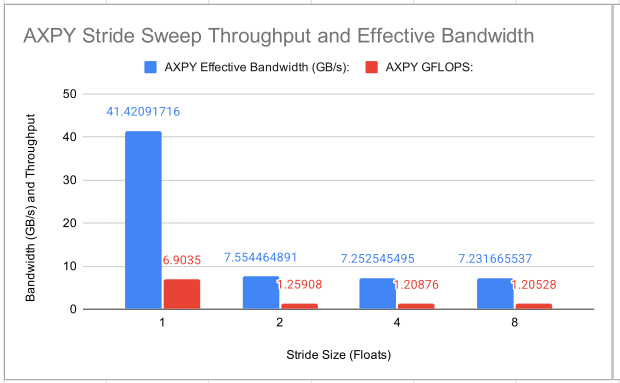


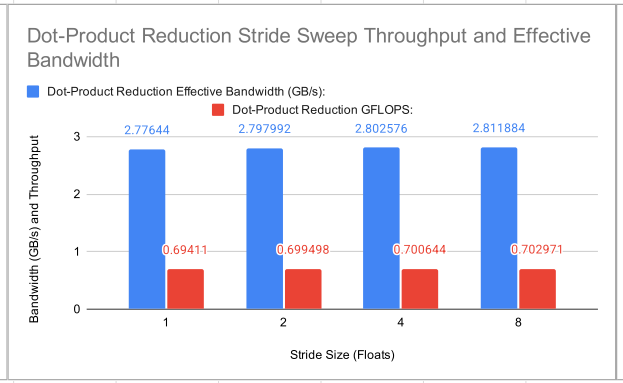


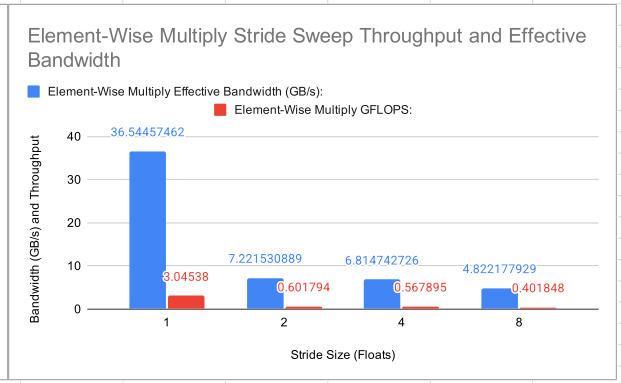
Kernel:	Aligned GFLOPS:	Misaligned GFLOPS:	% Drop:
AXPY:	6.90838	6.96454	-0.8063705571
Dot-Product Reduction:	0.692842	0.690639	0.3189799591
Element-Wise Multiply:	3.0477	3.07783	-0.9789364585



Stride Size:	AXPY Effective Bandwidth (GB/s):	AXPY GFLOPS:	AXPY HW Arithmetic Intensity (FLOPS/Byte):	Dot-Product Reduction Effective Bandwidth (GB/s):	Dot-Product Reduction GFLOPS:	Dot-Product Reduction HW Arithmetic Intensity (FLOPS/Byte):	Element-Wise Multiply Effective Bandwidth (GB/s):	Element-Wise Multiply GFLOPS:	Element-Wise Multiply HW Arithmetic Intensity (FLOPS/Byte):
1	41.42091716	6.9035	0.166667	2.77644	0.69411	0.25	36.54457462	3.04538	0.0833333
2	7.554464891	1.25908	0.166667	2.797992	0.699498	0.25	7.221530889	0.601794	0.0833333
4	7.252545495	1.20876	0.166667	2.802576	0.700644	0.25	6.814742726	0.567895	0.0833333
8	7.231665537	1.20528	0.166667	2.811884	0.702971	0.25	4.822177929	0.401848	0.0833333

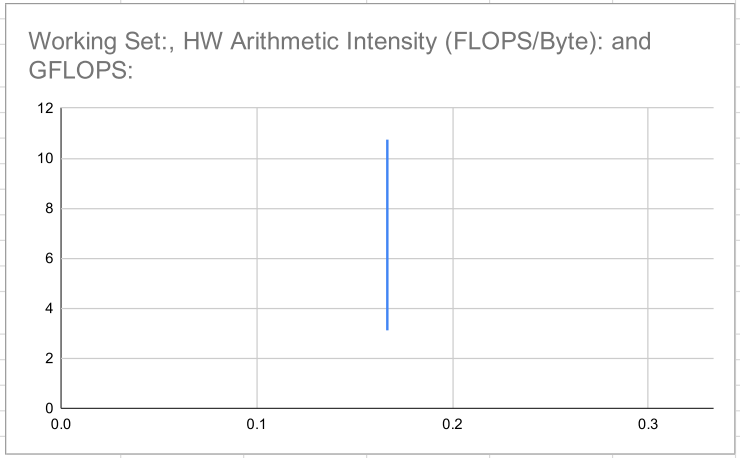




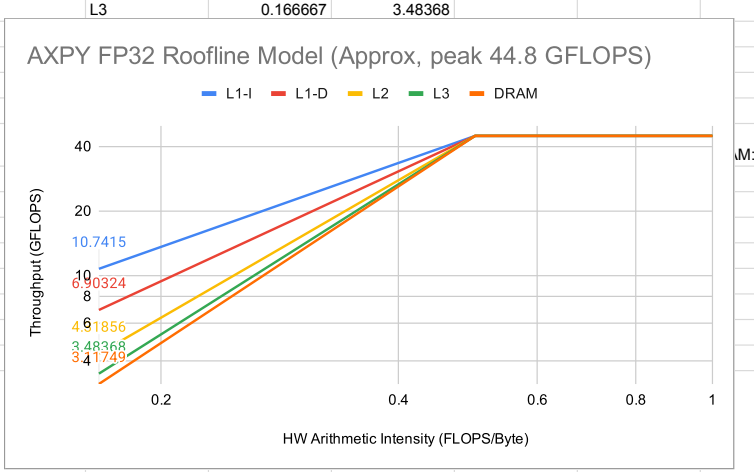


(AVX-512, 512bit)	Lane Count:	AXPY Arithmetic Intensity (FLOPS/Byte):	AXPY Operation Cost (FLOPS/element):	Dot-Product Reduction Arithmetic Intensity (FLOPS/Byte):	Dot-Product Reduction Operation Cost (FLOPS/element):	Element-Wise Multiply Arithmetic Intensity (FLOPS/Byte):	Element-Wise Multiply Operation Cost (FLOPS/element):
FP32	16	0.166667	2	0.25	2	0.0833333	1
FP64	8	0.0833333	2	0.125	2	0.0416667	1

AXPY:	Working Set:	HW Arithmetic Intensity (FLOPS/Byte):	GFLOPS:	Effective Bandwidth (GB/s):	Dot-Product Reduction:	HW Arithmetic Intensity (FLOPS/Byte):	GFLOPS:	Effective Bandwidth (GB/s):	Element-Wise Multiplication:	HW Arithmetic Intensity (FLOPS/Byte):	GFLOPS:	Effective Bandwidth (GB/s):
L1-I	4096	0.166667	10.7415	64.4488711		0.249998	0.70593	2.82374259		0.0833333	5.38983	64.67798587
L1-D	9216	0.166667	6.90324	41.41935716		0.2499999	0.699699	2.79879712		0.0833333	3.02779	36.33349453
L2	262144	0.166667	4.31856	25.91130818		0.25	0.690066	2.760264		0.0833333	1.50797	18.09564724
L3	1572864	0.166667	3.48368	20.9020382		0.25	0.689951	2.759804		0.0833333	1.18079	14.16948567
DRAM	16777216	0.166667	3.11749	18.70490259		0.25	0.689777	2.759108		0.0833333	1.12014	13.44168538



AXPY:	HW Arithmetic intensity:	GFLOPS:
L1-I	0.166667	10.7415
L1-D	0.166667	6.90324
L2	0.166667	4.31856
L3	0.166667	3.48368



M:
0
3.11749
44.8
44.8

Roofline AXPY approx

(AVX-512, 512bit)	Lane Count:	AXPY Arithmetic Intensity (FLOPS/Byte):	AXPY Operation Cost (FLOPS/element):
FP32	16	0.166667	2
FP64	8	0.0833333	2

Peak Theoretical GFLOPS (FP32, at 1.40GHz, single core, AVX-512, i5 1135G7): 1 FMA per core * 16 lanes per vector * 2 FLOPS/element * 1.40GHz clockspeed = 44.8 GFLOPS

Peak Theoretical GFLOPS (FP64, at 1.40GHz, single core, AVX-512, i5 1135G7): 1 FMA per core * 8 lanes per vector * 2 FLOPS/element * 1.40GHz clockspeed = 22.4 GFLOPS

Bytes per 512-bit vector: 64 bytes

HW arithmetic intensity for FP32 max:

HW arithmetic intensity for FP64 max:

hw arithmetic intensity = total flops / total bytes moved	total bytes moved: = 64 bytes per 512-bit vector, how many vectors? 2 flops per element, 4-8 bytes per element	
FP32: 44800000000 flops / 89600000000 = 0.5 FLOPS/Byte	for fp32: 16 elements per vector * 2 flops per element = 32 flops per vector	44.8 gflops / 32 flops per vector = 1.4 billion vectors, * 64 bytes per vector = 89.6 billion bytes
FP64: 22400000000 flops / 89600000000 = 0.25 FLOPS/Byte	for fp64: 8 elements per vector * 2 flops per element = 16 flops per vector	22.4 gflops / 16 flops per vector = 1.4 billion vectors, *64 bytes per vector = 89.6 billion bytes